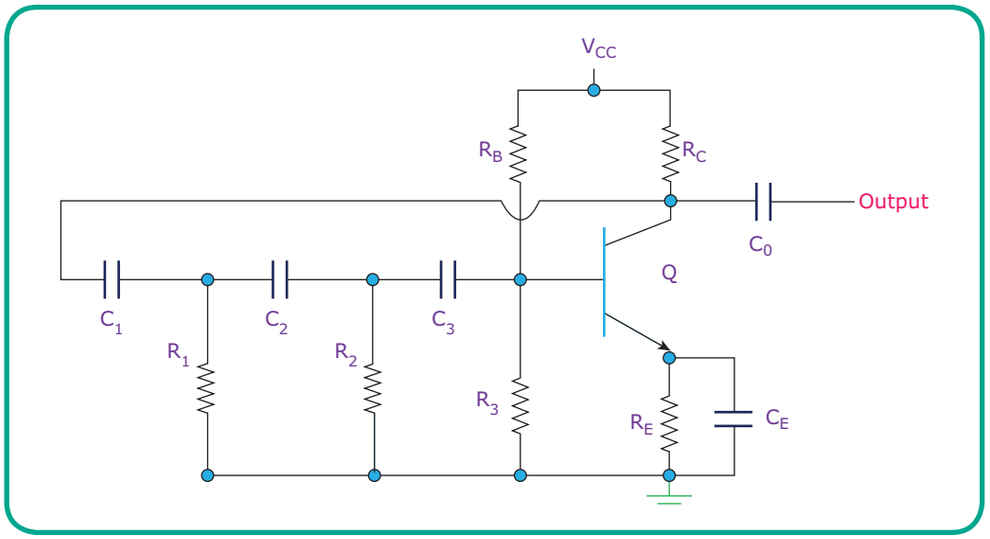


Figure 7.7 **RC Phase Shift Oscillator Using BJT**

Working Principle: When the circuit is switched 'ON', random variation in the DC power supply or noise variation in the transistor oscillator occurs. This can be amplified by the transistor. The output 'E_O' of the amplifier is feedback through the RC network. This network produces a phase shift of 180⁰ and a voltage E_i appears at its output is applied to the transistor amplifier. Obviously, the feedback fraction $m=E_i/E_0$, if the feedback phase is correct, a phase shift of 180⁰ is produced by the RC network. As a result, the phase shift around the entire loop is 360⁰.

Advantages

1. Does not require transformer or inductor
2. It can produce very low frequency signal
3. The circuit provides good frequency stability

Disadvantages

1. It gives low power output
2. It is difficult to start oscillation as the feedback signal is small

7.5.3. Crystal Oscillator

The transistor crystal oscillator resembles Colpitts oscillator modified to act as crystal oscillator. The only change is the addition of the crystal (Y) in the feedback network. It replaces the LC resonant circuit.


W. G. Cady

History: Piezoelectricity was discovered by Jacques and Pierre Curie in 1880. Paul Langevin first investigated quartz resonators for use in sonar during World War I. The first crystal controlled oscillator using a crystal of Rochelle salt was built in 1917 and patented in 1918 by Alexander M. Nicholson at Bell Lab, although his priority was disputed by Walter Guyton Cady. Cady built first quartz crystal oscillator in 1921.

The crystal is a thin slice of piezoelectric material, such as quartz, tourmaline and Rochelle salt, which exhibits a property called piezoelectric effect. It means the crystal reacts to any mechanical stress by producing electric charge, in

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